

PROFESSIONAL SUMMARY

- 10+ years of combined multi-disciplinary experience in industry and academia as an applied data scientist, with expertise in optimization, machine learning, deep learning, and reinforcement learning.
- Recognized thought leader driving multi-million-dollar business impact across Energy, Real Estate, and Manufacturing domains; technical lead on several AI solutions in Walmart's Energy Platform.
- Inventor of 2 US patents, author of 6 journals, 10 conference papers, and 1 book chapter with 500+ citations; champion mentor and cross-functional leader across product, engineering, and business teams.

WORK EXPERIENCE

Walmart Inc.

Bentonville, AR, USA

Staff Data Scientist

Nov. 2024 – Present

- Served as technical lead and primary DS owner for Walmart's Energy Transformation initiatives, directing end-to-end solution delivery for solar/battery (DG), generator resiliency, and EV charging pricing; mentored 2 junior associates in energy rate forecasting, EV demand modeling, and power outage analytics.
- DG Site Selection Framework: Deployed multi-cloud API integrating solar, battery dispatch, and financial IRR models; automated 10,000 analyst hours annually, avoided \$200K/site investment risk, and enabled executive-level reporting for 1,000+ site pipeline supporting \$300M+ energy cost savings goal by FY31; work leading to US patent "Systems and Methods for Photovoltaic Power Environment Deployment".
- Generation & Resiliency: Developed a comprehensive site-recommendation framework integrating outage history, weather-hazard composite risk score, and geo-spatial proximity analytics — supporting \$200M annual savings potential for generator deployment decisions.
- EV Pricing Engine: Deployed TOU/hourly pricing recommendation API via MLOps using competitor pricing, utility cost, and hourly EV demand forecasting; solution to be leveraged by 300 sites in scope by FY27 end; each \$0.005 pricing is worth \$15M in revenue through FY31.

Senior Data Scientist

May 2022 – Oct. 2024

- Solar Site Selection: Built end-to-end data pipeline and PVLib-based solar generation model; delivered business insights presented at Energy Transformation board meeting — directly pivotal for approving upcoming solar projects; paper awarded Best Paper at SparkTech 2024 among 1,300+ submissions.
- Solar Image Segmentation: Developed semantic segmentation model with U-Net architecture using 100 satellite images achieving 87% accuracy for determining solar; built rooftop obstacle detection and optimal panel placement models for accurate estimation of solar energy potential at Walmart sites.
- Energy Arbitrage with Battery: Formulated and solved the behind-the-meter battery arbitrage and peak-demand shaving problem using Linear Programming; achieved 8X improvement in optimization computation time via matrix formulation and HiGHS solver.
- Filed US Patent "Systems and Methods for Simulation Optimization of Production Networks"; published 1 external conference paper on Energy Efficiency benchmarking and 2 Medium tech blogs on genetic algorithms and geo-spatial analytics.

Data Scientist

Jun. 2020 – Apr. 2022

- Served as optimization SME across Real Estate and Manufacturing domain; led Department Space Optimization (Linear Programming / R Shiny MVP), Market Optimization for New Store Site Selection (Genetic Algorithm), and Remodel Task Schedule (NSGA-2) models.
- Formulated multi-objective genetic algorithm optimization to solve the production scheduling of a beef manufacturing plant improving schedule adherence by 20% and capacity utilization by 10%; model became foundation of approved US patent.
- Audited and streamlined 7,000 lines of code across 43 Python scripts; migrated Alteryx workflows to Python, advancing MLOps maturity and significantly accelerating product enhancement cycles.

Data Analyst Intern

Jun. 2019 – Aug. 2019

- Developed regression and optimization models in R for respectively estimating & allocating maintenance budget for HVAC & refrigeration equipment in stores and deployed a POC app using Alteryx.

University of Arkansas

Fayetteville, AR, USA

Graduate Research and Teaching Assistant

Jan. 2016 – May 2020

- Published 6 journal papers, 6 conference papers, and 1 book chapter with research findings in the field of Optimization, Statistical Signal Processing, Deep Reinforcement Learning, and Wireless Communication with simulations conducted in Matlab & Python.
- Dissertation Topic: Presented low-latency anomaly detection algorithms for smart grid (false data injection) and wind turbine fault detection.
- Deep Reinforcement Learning: Developed a DRL-based approach with Deep Deterministic Policy Gradient (DDPG) for dynamic computation offloading and resource allocation for mobile edge computing systems.
- Optimized Scheduling: Designed multiple methods using optimal stopping theory and Markov decision process (MDP) to solve information scheduling in wireless networks while balancing the tradeoff among information delay, energy efficiency, and system cost.

EDUCATION

Doctor of Philosophy (Ph.D.) in Electrical Engineering

May 2020

University of Arkansas

Fayetteville, AR, USA

Dissertation: Low Latency Anomaly Detection with Imperfect Models

Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering

Jul. 2014

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

Thesis: Spatio-Temporal Feature Extraction Scheme for Human Action Recognition

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL, Matlab
- **ML/DL Libraries:** Scikit-learn, Keras, TensorFlow, PyTorch, PySpark, MLflow
- **Optimization:** Linear Programming, MILP, Genetic Algorithm, Reinforcement Learning
- **Deployment:** MLOps, Git, CI/CD, Docker
- **Tools:** Streamlit, R Shiny, Tableau, Alteryx, Jira, L^AT_EX

PATENTS & SELECTED PUBLICATIONS

- [Patent] **S. Nath**, V. Patel, et al., “Systems and Methods for Photovoltaic Power Environment Deployment”, US Patent Application No. 19/539,424, Feb 2026 (*Filed*).
- [Patent] R. Angadala, **S. Nath**, et al., “Systems and Methods for Simulation Optimization of Production Networks”, US Patent Application No. 18/428,941, Jul 2025.
- **S. Nath** and J. Wu, “Deep Reinforcement Learning for Dynamic Computation Offloading and Resource Allocation in Cache-assisted Mobile Edge Computing Systems”, *Intelligent and Converged Networks*, 2020.
- **S. Nath**, J. Wu, et al., “Low Latency Bearing Fault Detection of Direct-drive Wind Turbines Using Stator Current”, *IEEE Access*, vol. 8, 2020.
- **S. Nath**, I. Akingeneye, et al., “Quickest Detection of False Data Injection Attacks in Smart Grid with Dynamic Models”, *IEEE JESTPE*, vol. 10(1), 2019.

AWARDS & CERTIFICATIONS

- Best Paper Award at Walmart SparkTech Summit Jul. 2024
- Making a Difference Award at Walmart for going above & beyond Aug. 2024, Dec. 2024, Oct. 2025
- Completed Deep Learning Specialization in Coursera Oct. 2023
- Completed Dale Carnegie: Skills for Success Training Nov. 2021
- [In Progress] Gen AI - LLM (DeepLearning.AI), Agentic AI (Vanderbilt) 2026